Part A – Exploratory Data Analysis (EDA) and Insights

a) Summary of Company_X_Audience and Company_X_ShowData datasets

There are two datasets:

1. Company_X_Audience
   This dataset gives an overall view of the audience purchasing behaviors, engagement during the event and includes vital demographics.

2. Company_X_ShowData
   This dataset presents details of the events including the date, total revenue, sales and weather conditions.

Although the data collection timeframe is not mentioned we could derive that the Company_X_Audience dataset contains records from 2025, while the Company_X_ShowData dataset contains records from 2025 to February 2026.

Dataset Dimensions

Dataset	Columns	Rows
Company_X_Audience	13	800
Company_X_Show	10	200

Data Types

1. Company_X_Audience

- Categorical:
  - Boolean: Repeat_Visit
  - Nominal: Gender, Country
  - Ordinal: Seating_Region

- Numerical:
    - Continous (Integer): Age, Ticket_Price
    - Continous(float): Merchandise_Spend, Drink_Spend, Satisfaction_Score, Recommendation_Likelihood
    - Discrete: Num_Tickets

- Date:
    - Visit_Date

2. Company_X_ShowData

- Categorical:
    - Boolean: Guest_Performance
    - Nominal: Date_Type, Region

- Numerical (Continous): Attendance, Avg_Ticket_Price, Avg_Merch_Sales, Revenue, Weather_Score

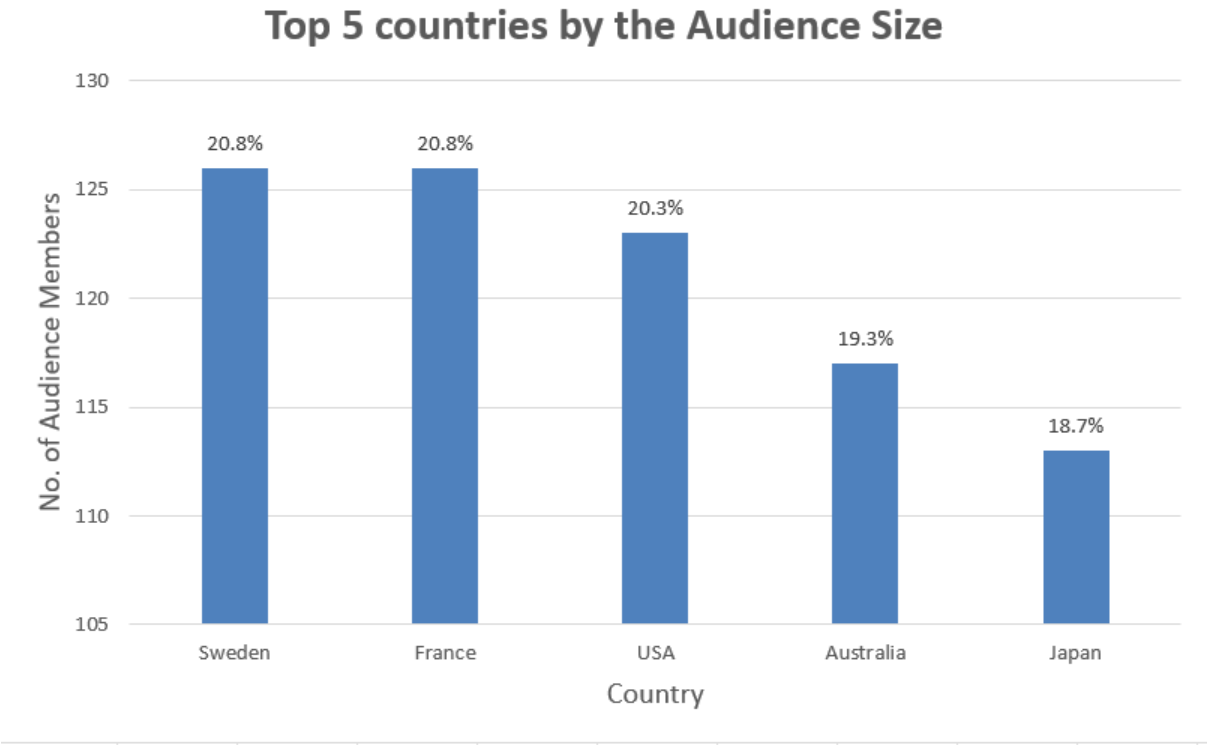
- Date:
    - Show_Date

Missing Values and Data Quality

To ensure data quality of the two datasets data cleaning was processed to check for duplicate records or missing values. There were no duplicate records or missing values were found in either dataset. But during data validation a non-numerical ('y') was identified in the Attendance feature in Company_X_ShowData dataset. This was handles by treating as a data error and replacing with a blank value since deleting the record could lose vital information in other variables in the records. However, all the other features matched their expected data type.

b) Top Five Countries contributing the Highest Audience

The top five countries contributing the highest number of audience were: Sweden, France, USA, Australia and Japan.



Observation 1:

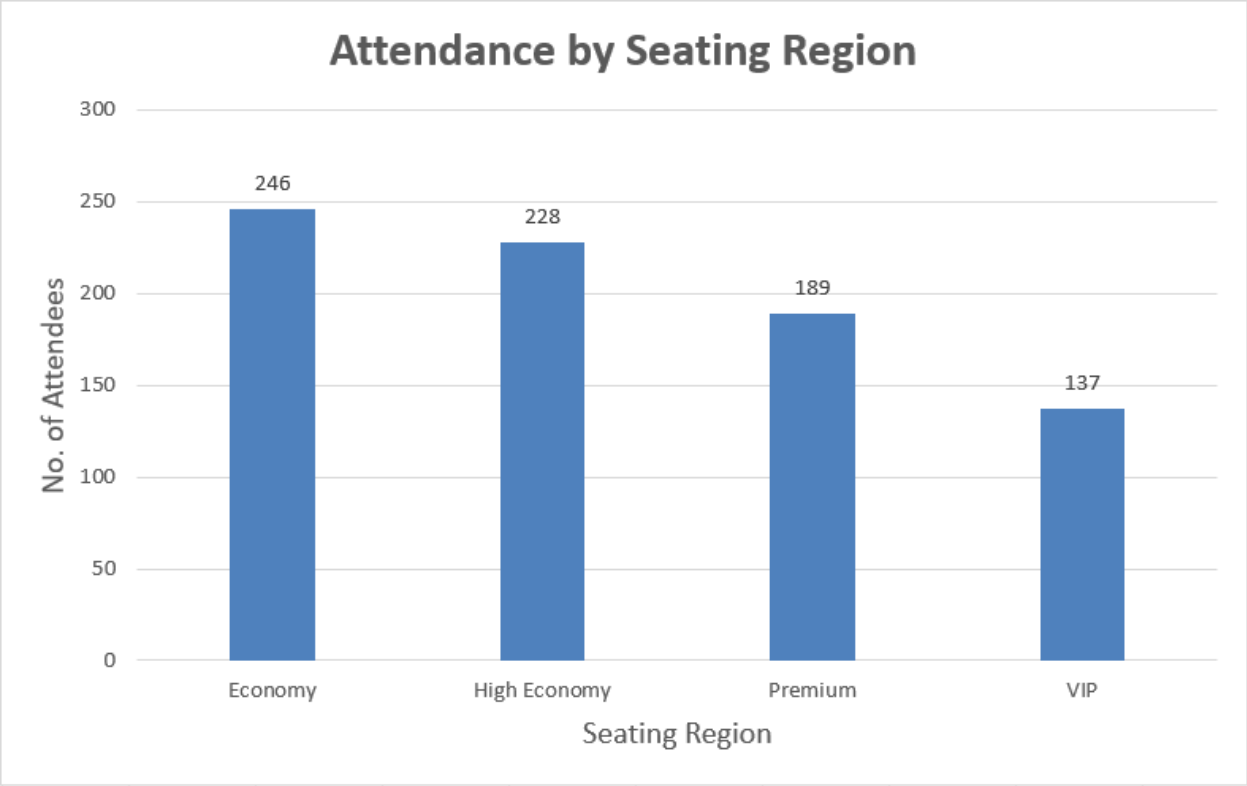
Sweden and France jointly contribute the highest audience with approximately 20.8% each of the total audience among the top five countries. However, USA is ranked third with an approximation of 20.3%, which is a relatively small difference of 0.5% with Sweden and France. So, audience distribution is fairly balanced among the top three countries. Therefore, Sweden, France and USA are among the leading markets.

Observation 2:

Australia and Japan have the lowest audience among the top five countries contributing approximately 19.3% and 18.7% respectively. However, Japan has 2.1% and Australia has 1.5% lower engagement levels comparing with Sweden and France which displays a moderate distribution among the top five countries.

c) Attendance Across Different Seating Regions

To analyze the attendance across different seating regions the Customer_ID and Seating_Region features from the Company_X_Audience dataset is used. Then the data is aggregated by customer count in each seating region. For visualization column bar chart is used because Seating_Region is a categorical feature so it is effective to show the distribution across different seating regions and identify patterns and disparities in audience.



Economy has the highest attendance contributing 31% of the total attendance, indicating most people prefer selecting lower cost options. However, it's only 2% higher than High Economy which contributes 29% of the total audience. So, the demand is relatively distributed at the lower tier seatings.

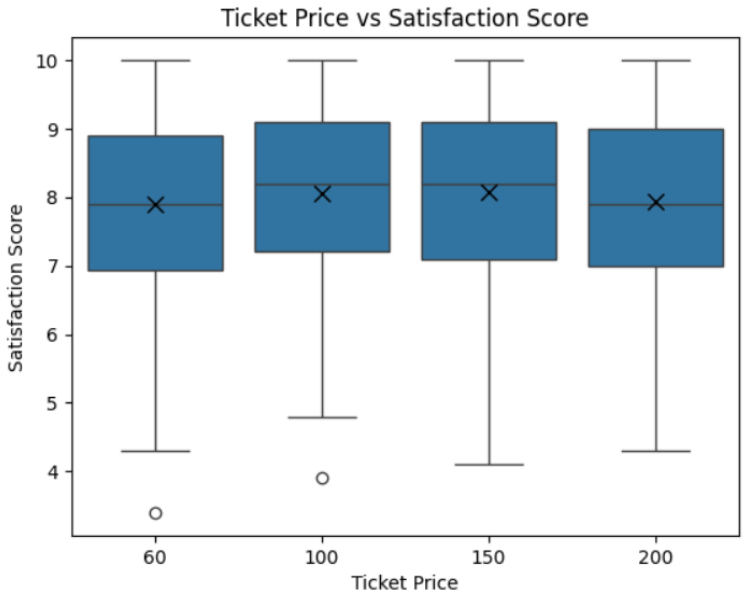
Since, High Economy is slightly closer to Economy region this indicates attendees are willing to buy ticket bit higher for slightly better comfort and quality. This shows a balanced preference between budget and value-for-many options.

There is a steady decline from lower tier to higher tier seating regions. Premium seating contributes 24% and VIP contributes 17% of the total audience displaying lower attendance comparing with lower tier seatings.

Although, VIP has the lowest attendance it has a proportionate amount. The disparity between the four seating regions doesn't have an extreme imbalance. So, there is not poorly showing attendance in any seating regions.

d) Relationship between Ticket Price and Satisfaction Score

To visualize the relationship between Ticket Price and Satisfaction Score a box plot is used as it effectively shows the distribution, median, range, interquartile range and potential outliers across different ticket price categories.



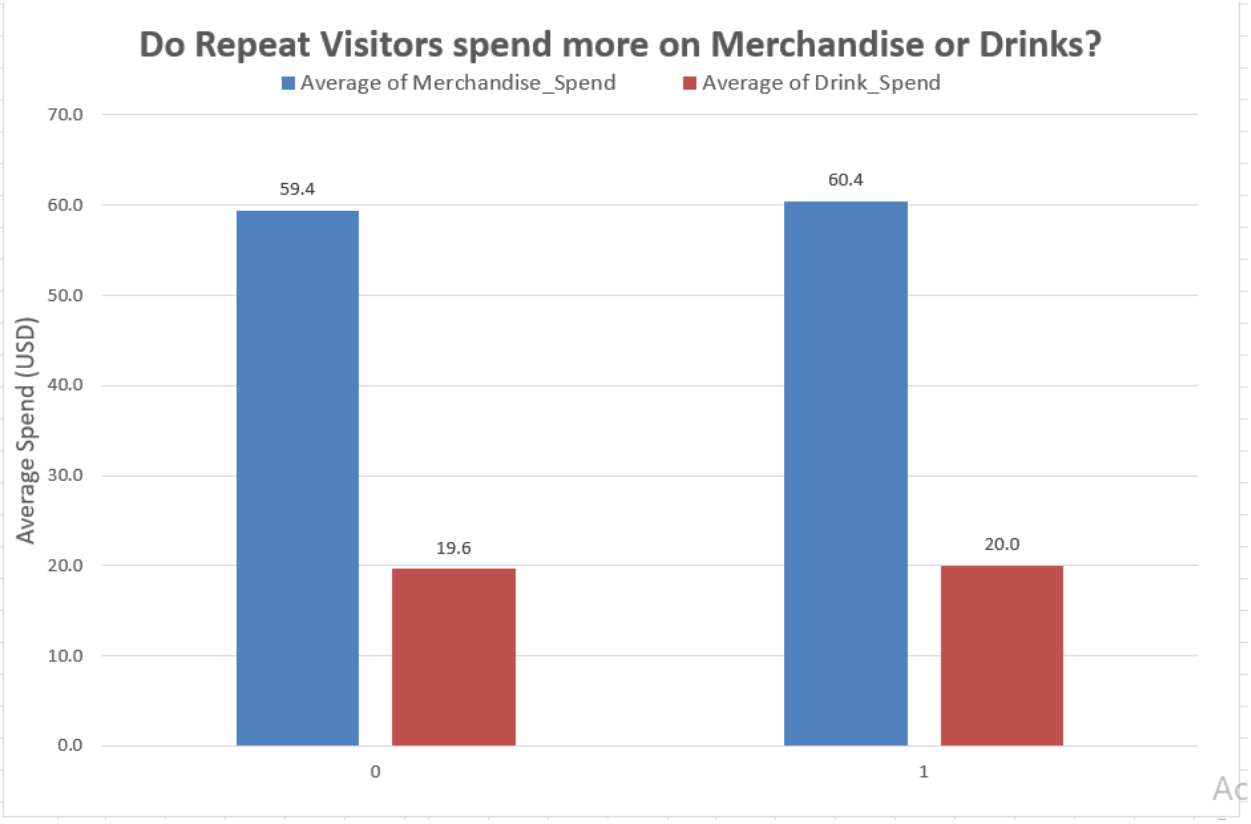
According to the visualization, the interquartile range of all four ticket prices have very slight difference displaying almost similar box sizes falling roughly between a satisfaction score of seven and nine. This presents a consistent variability in satisfaction score across all ticket prices. The median value in all four ticket prices is around eight and the mean for all ticket prices is close to 7.5 and 8. Indicating consistent high satisfaction across in all ticket price categories. There are few outliers in lower ticket prices (60, 100) which shows some customers had low satisfaction in the lower ticket prices. Moreover, all ticket prices have longer lower whiskers showing a left skewness where few poor ratings pull down the distribution slightly. However, the skewness is not strong indicating that most of the customers have high satisfaction but relatively few customers had dissatisfaction. Therefore, there is no strong relationship with ticket price and customer satisfaction.

Key Insights:

1. Customer satisfaction is high in all ticket price categories so decreasing or increasing the ticket price does not increase the customer's satisfaction. Therefore, Company X can adjust price strategies in tickets without risking the satisfaction of the customers.
2. Since, there is a small amount of dissatisfaction from customer from all ticket price categories, Company X should identify the reasons and address to improve the satisfaction.

e) Analyzing whether Repeat Visits tend spend more on Merchandise or Drinks

To analyze whether repeat visits spend more on merchandise or drinks, the Repeat_Visit, Merchandise_Spend and Drink_Spend features were encountered from Company_X_Audience dataset. Then the data is aggregated by repeat_visit and averaging the merchandise spend and drink spend for repeat and non-repeat visits. A column chart is used for visualization to effectively compare the spend between repeat visits and non-repeat visits.



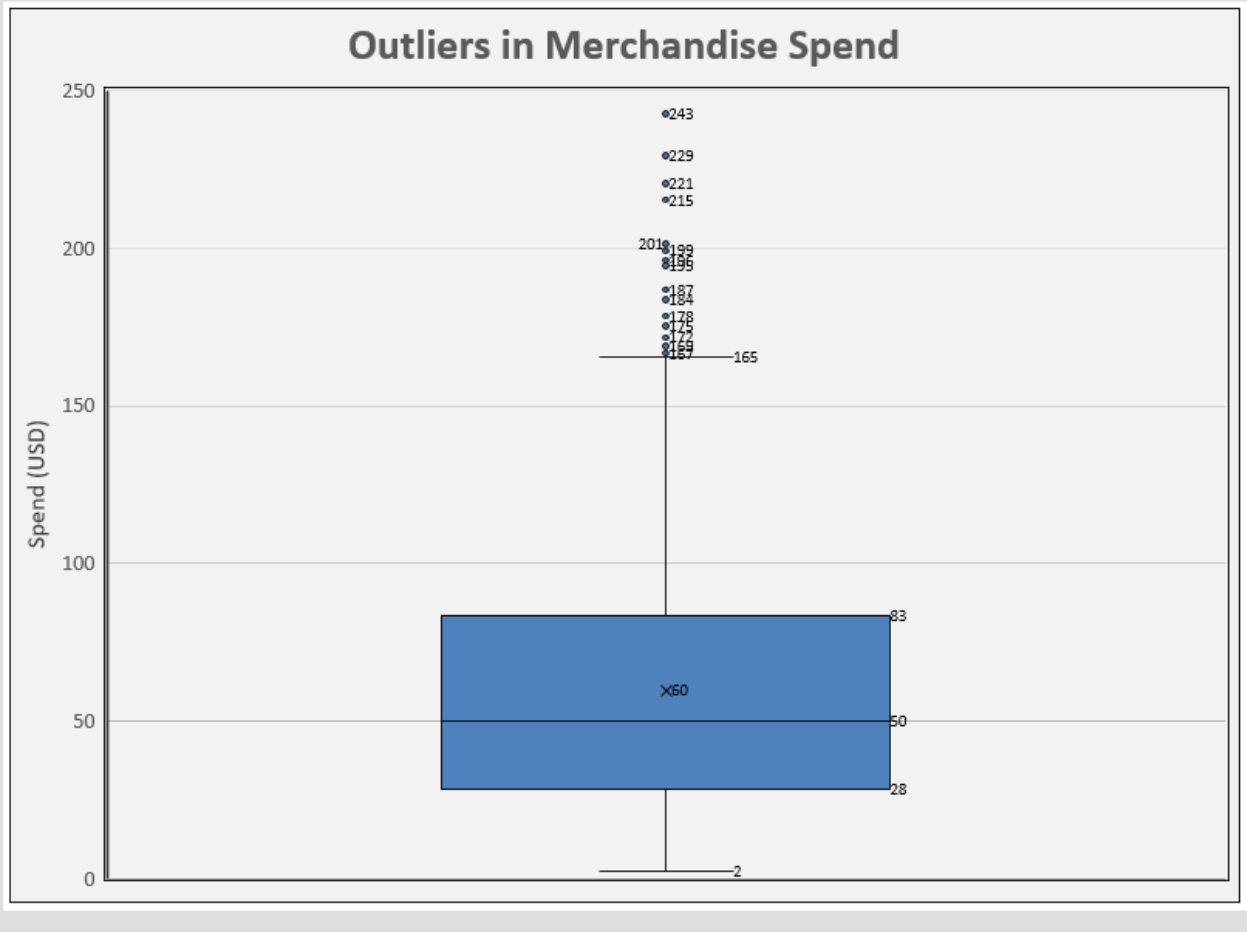
Repeat visitors spend averagely USD 60.4 on merchandise and USD 20.0 on drinks whereas, non-repeat visitors spend averagely USD 59.4 on merchandise and USD 19.6 on drinks. This shows, repeat visitors spend slightly higher than non-repeat visitors but not a significant difference between repeat visits and non-repeat visits.

Repeat visitors tend to spend more on merchandise than drinks indicating a higher attraction on merchandise.

However, in contrast with non-repeat visitors, they too have almost similar interest purchasing merchandise rather than drinks meaning repeat visits do no affect spending behavior.

f) Identifying potential outliers or Anomalies for high spending and satisfaction score

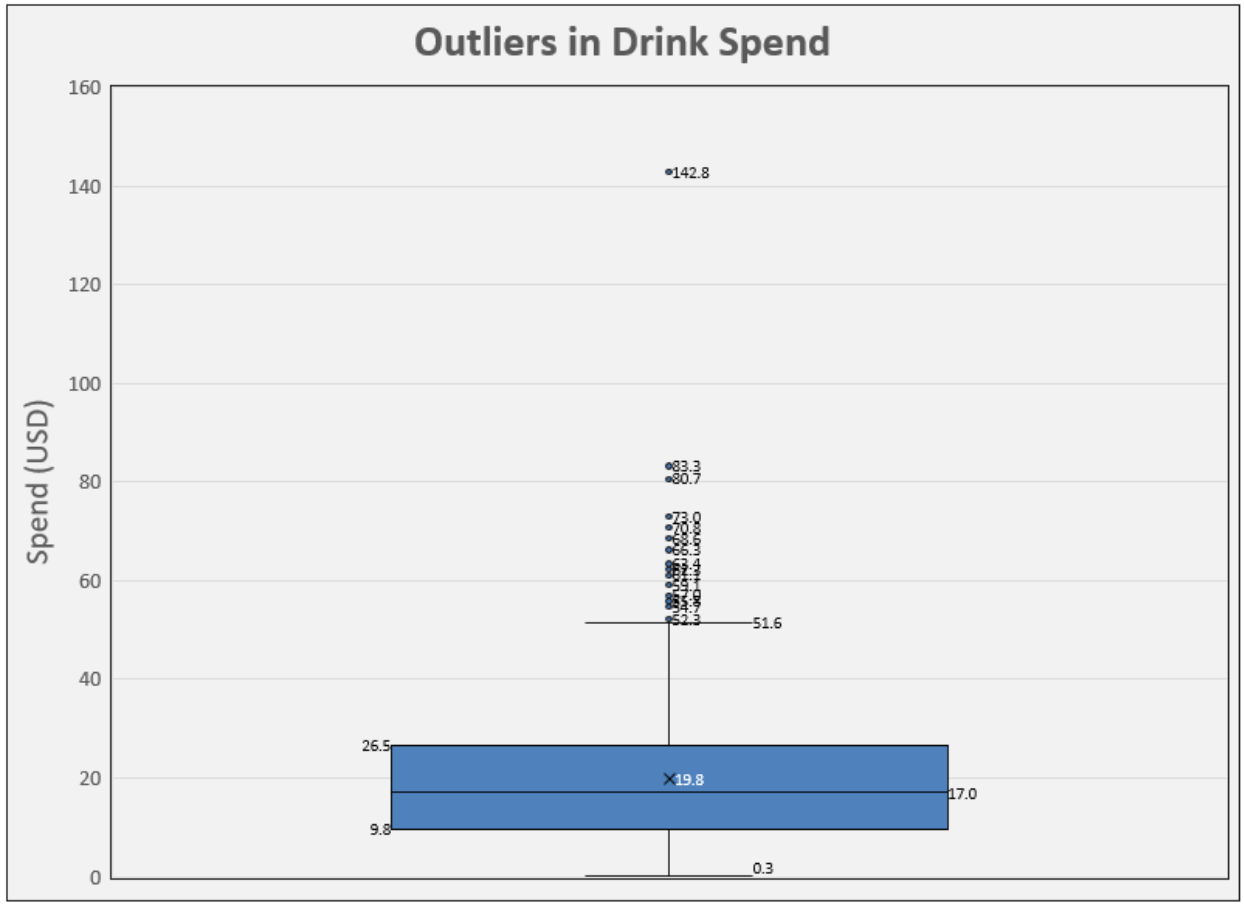
Outliers in Merchandise Spend



Mean is slightly higher than median. The upward whisker is longer and has few outliers which leads to pulling the mean upward. So, the data is positively skewed.

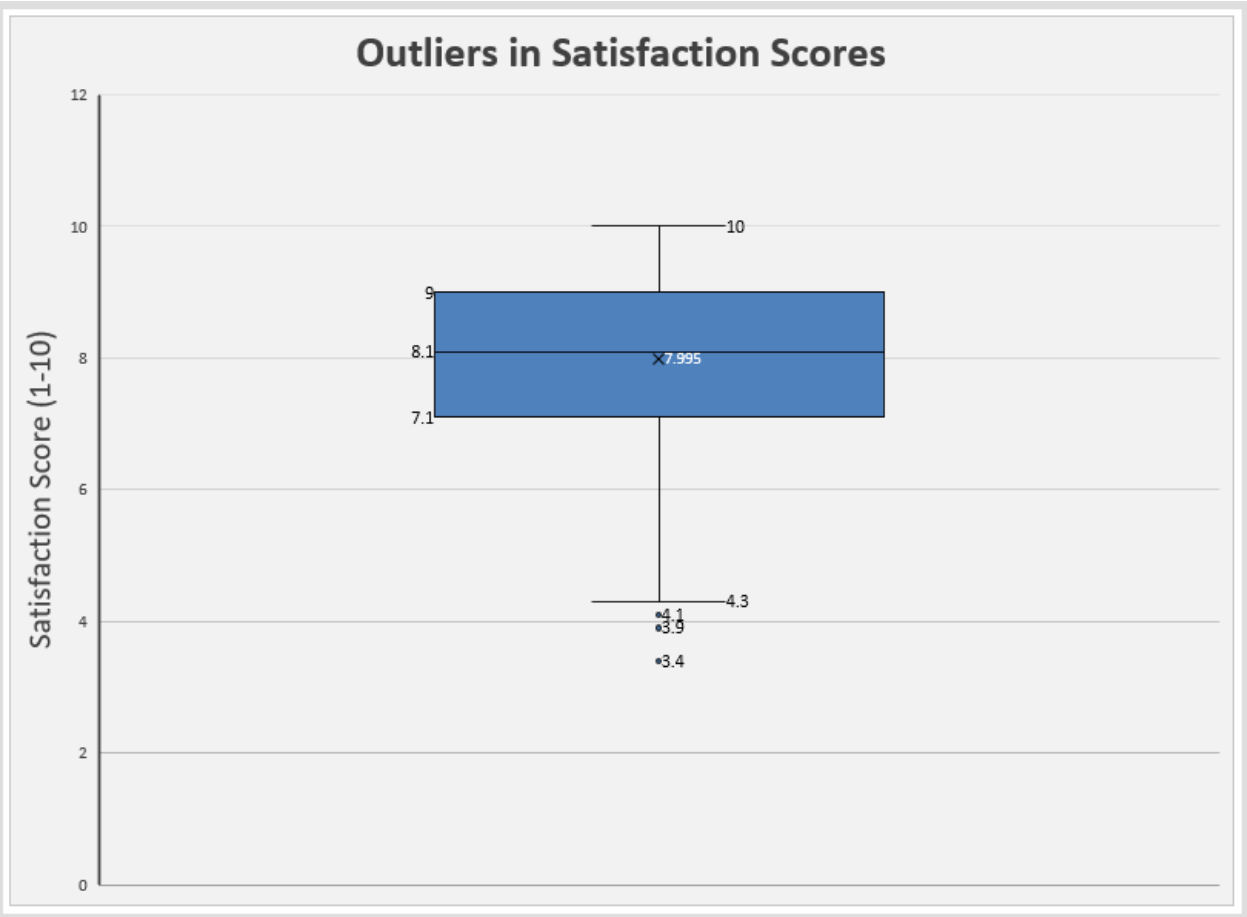
Lower whisker extends towards $2 and upper whisker goes until $165 which indicates typical customers buy around this range and beyond this is unusual. There few outliers and some are extreme which shows a small group of customers spends higher than the normal. This could be VIP or abnormal spending behavior. So, these people contribute disproportionately to the revenue indicating presence of high-value customers.

Outliers in Drink Spend



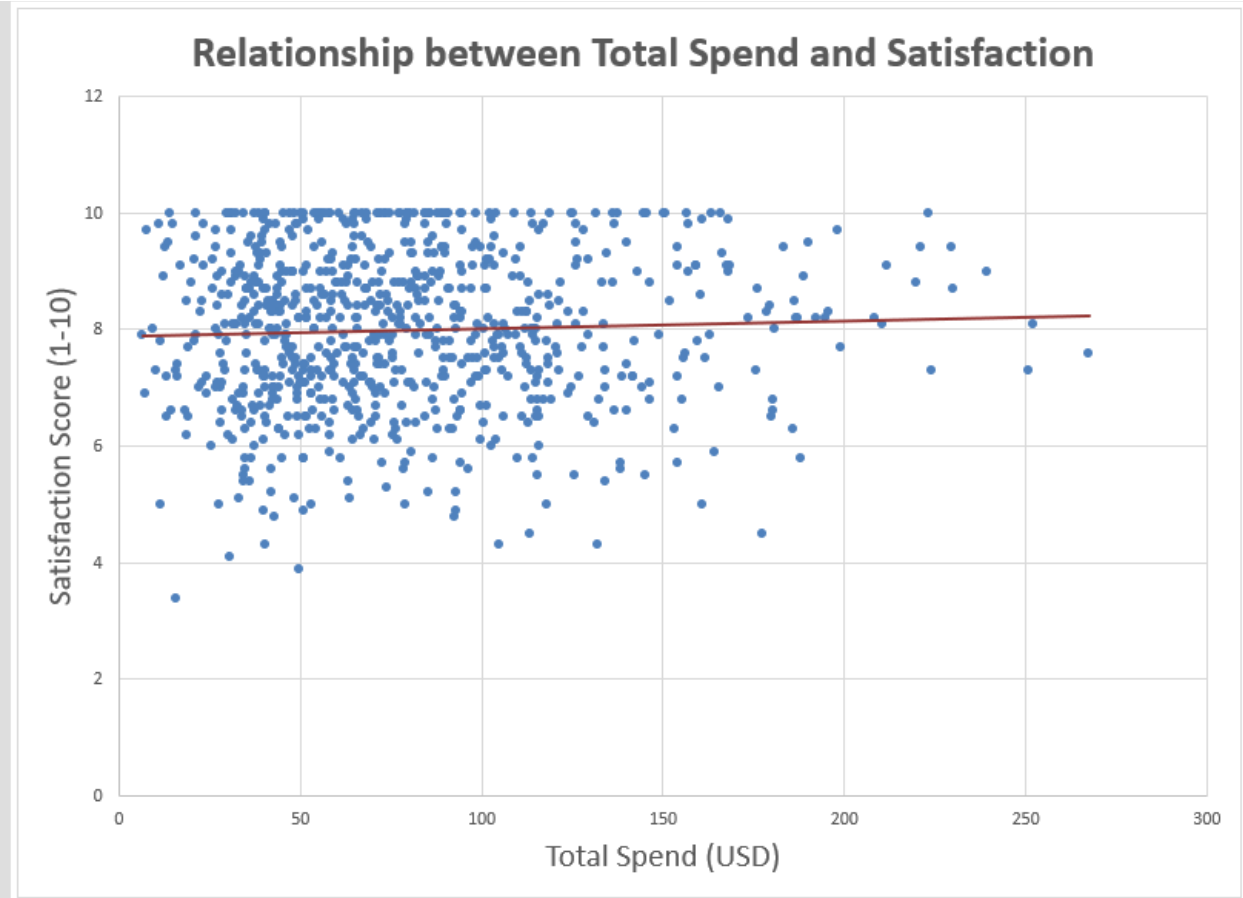
The median is around $17. The mean is slightly higher than median indicating a right skewed distribution. The 50% visitors spend between $10 and $26.5 showing low to moderate drinkers but typical drinker stays below $52. There are many outliers above the upper whisker presenting there are high drink spenders which could be in large groups. Mostly the customers are concentrated at lower values showing low drinkers but there are high spenders that contribute to the revenue.

Outliers in Satisfaction



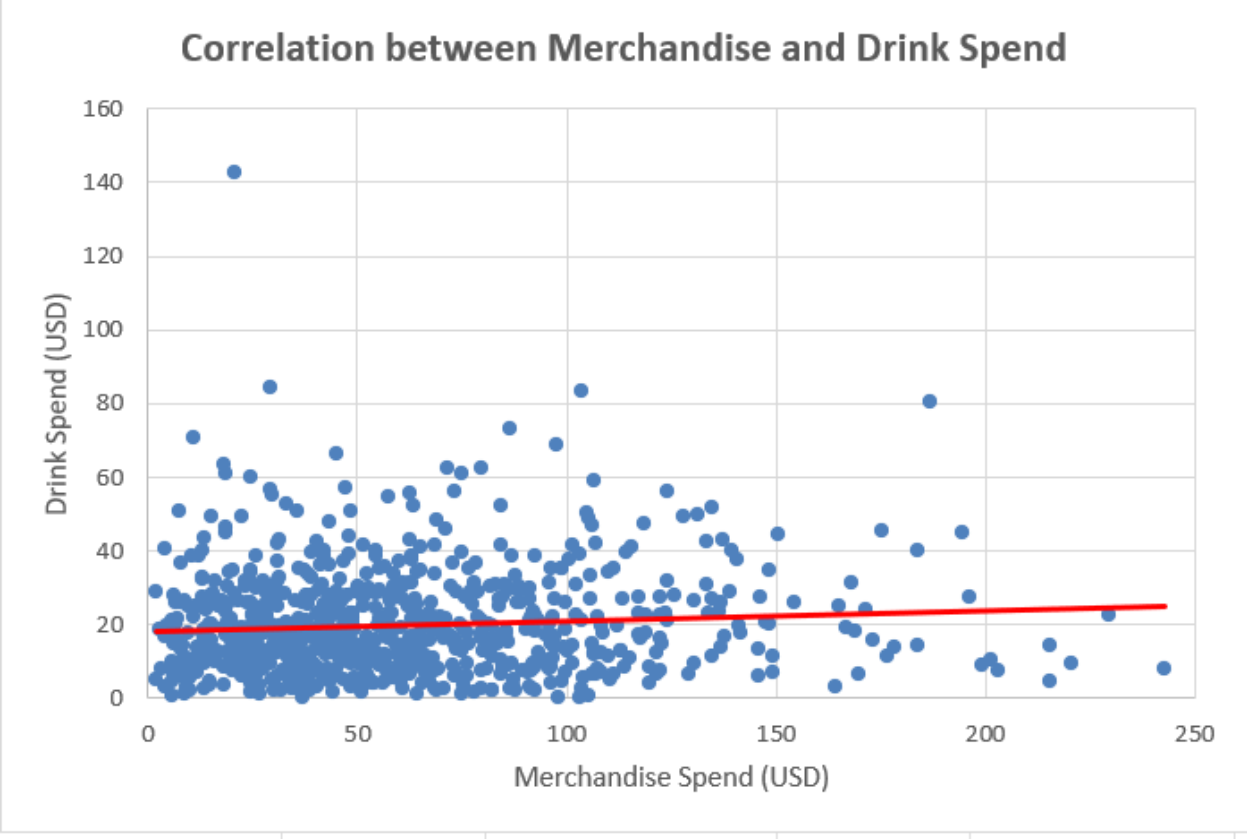
The median satisfaction score is approximately 8.1, with the majority falling between 7.1 and 9. This indicates that the product or event is usually well-received. Business meaning: The organization should concentrate on preserving this experience rather than making significant redesigns because the core offering is functioning properly.
This is a high level of customer approval because the majority of customers rank their satisfaction higher than 7. Business meaning: o It can be used in advertisements such as "Highly rated experience (8/10 average satisfaction)."

Relationship between Total Spend and Satisfaction Score



g) Correlation Among Ticket Price, Merchandise Spend, Drink Spend and Satisfaction

	Ticket_Price	Merchandise_Spend	Drink_Spend	Satisfaction_Score
Ticket_Price	1.000	-0.026	-0.055	0.017
Merchandise_Spend	-0.026	1.000	0.081	0.039
Drink_Spend	-0.055	0.081	1.000	0.021
Satisfaction_Score	0.017	0.039	0.021	1.000



Part B – Descriptive Statistics and Insights

a) Summary Statistics Table - Numerical Features

1)Company_X_Audience

Summary Statistics: Company_X_ShowData - Numerical Features

	Attendance	Avg_Ticket_Price	Avg_Merch_Sales	Revenue	Weather_Score
Mean	4407.13	121.56	34.62	688439.67	7.01
Median	4304.00	120.14	34.71	674747.32	7.15
Mode	3126.00	113.24	29.96	#N/A	10.00
Standard Deviation	1399.28	25.12	10.42	260405.41	2.02
Sample Variance	1957986.95	631.12	108.64	67810975000.22	4.08
Range	4919.00	142.88	62.35	1195828.45	9.00
Minimum	2045.00	48.75	2.58	243543.53	1.00
Maximum	6964.00	191.63	64.93	1439371.98	10.00
Sum	877018.00	24311.62	6924.98	137687934.93	1402.60
Count	199.00	200.00	200.00	200.00	200.00
Q1	3292.50	105.51	28.17	461509.82	5.70
Q3	5613.50	137.71	41.90	855444.14	8.60
IQR	2321.00	32.20	13.74	393934.32	2.90

2)Company_X_ShowData

Summary Statistics: Company_X_ShowData - Numerical Features

	Attendance	Avg_Ticket_Price	Avg_Merch_Sales	Revenue	Weather_Score
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Sum	877018.00	24311.62	6924.98	137687934.93	1402.60
Count	199.00	200.00	200.00	200.00	200.00
Q1	3292.50	105.51	28.17	461509.82	5.70
Q3	5613.50	137.71	41.90	855444.14	8.60
IQR	2321.00	32.20	13.74	393934.32	2.90

b) Summary Statistics Table - Categorical Features

1. Company_X_Audience

Statistical Summary: Company_X_Audience - Categorical Features

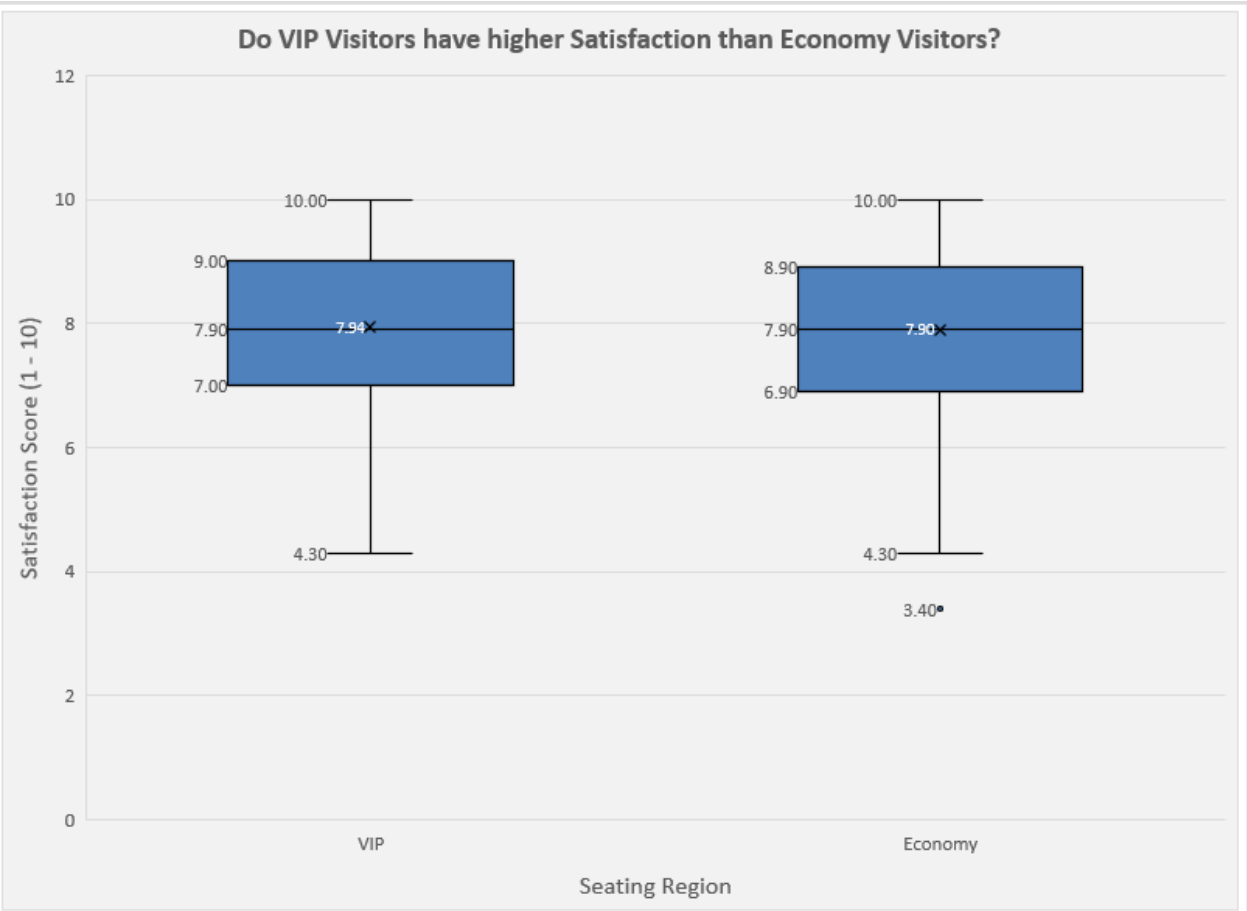
	Gender	Country	Seating_Region	Repeat_Visit
Count	800	800	800	800

2. Company_X_ShowData

Statistical Summary: Company_X_ShowData - Categorical Features

	Day_Type	Region	Gues_Performance
Count	200	200	200

c) Testing whether VIP region have a higher satisfaction than Economy Region



The satisfaction scores of VIP and Economy tourists were compared using a box plot. VIP guests have a slightly better mean satisfaction score (7.94) than Economy visitors (7.90). Nonetheless, there is a lot of overlap between the two groups' medians and general distributions. The distribution and range of values in both groups are comparable, however the Economy group exhibits a small outlier. As a result, there isn't much data to support the claim that VIP guests are substantially happier than Economy guests.

First insight: Higher satisfaction may not be justified by premium pricing

When compared to Economy visitors, VIP visitors do not exhibit appreciably better levels of satisfaction.

Higher satisfaction might not be justified by premium pricing. When compared to Economy region, VIP guests do not exhibit noticeably higher levels of satisfaction. VIP seating does not

provide customers with appreciably greater value. To support higher prices, improve VIP privileges (better service, incentives, exclusivity).

d) Interpreting the Engagement Value

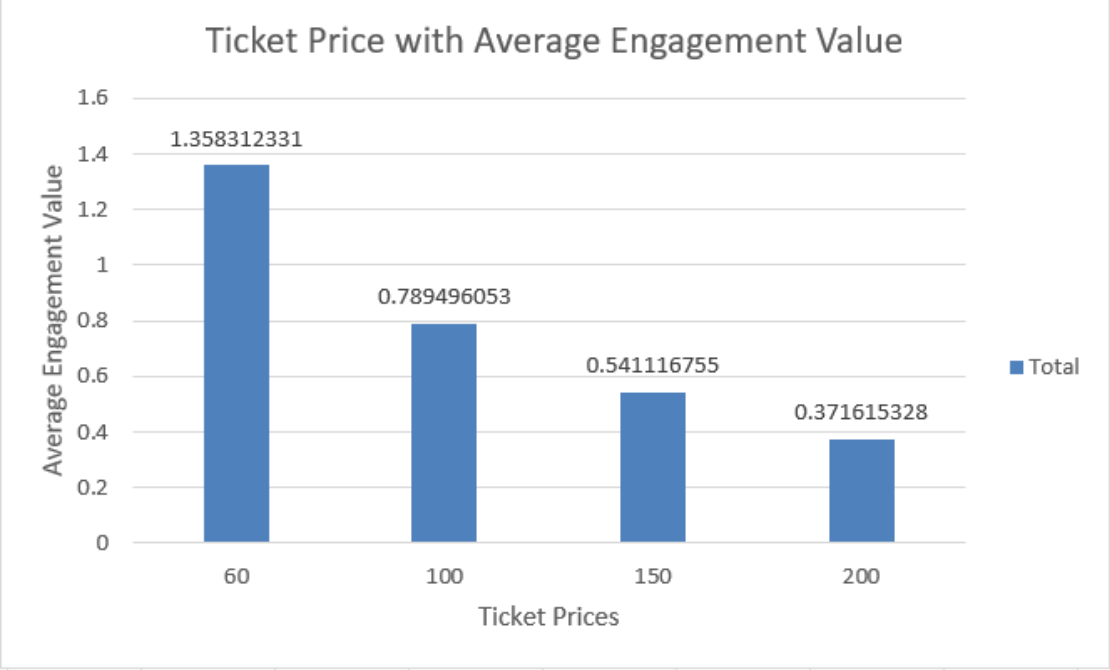
The ratio of the total extra money spent on beverages and merchandise to the cost of the ticket i s the Engagement_Value statistic.

It gives an indication of overall customer involvement by measuring the amount of money a cust omer spends after purchasing a ticket.

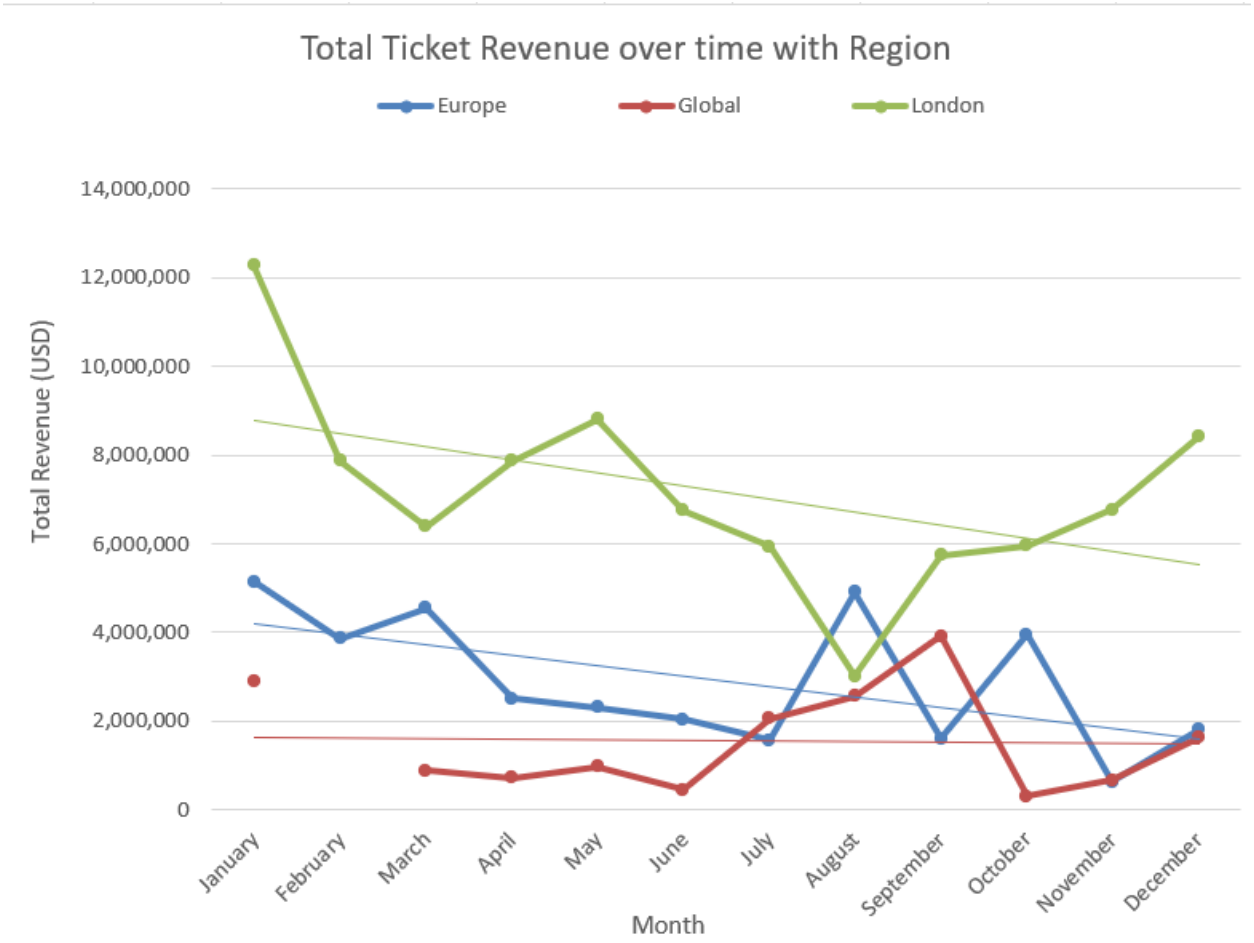
Customers are more actively involved when their Engagement_Value is higher since they spend more on extra goods in comparison to the cost of the ticket, whereas a lower value indicates les s engagement and less additional expenditure.

For Company X, this measure is crucial because it identifies high-value clients who make substantial contributions to total revenue outside of ticket sales. Additionally, it offers information about consumer spending patterns and the success of in-event sales tactics like beverage and item offerings.

Furthermore, the Engagement_Value can be used to compare different customer segments (e.g., seating regions or visitor types) to identify which groups are more likely to generate additional revenue. This enables the company to implement targeted promotions, improve product offerings, and optimise pricing strategies to increase overall profitability.


Ticket Price with Average Engagement Value
Average Engagement Value
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
1.358312331
0.789496053
0.541116755
0.371615328
60
100
150
200
Ticket Prices
Total

e) Total Ticket Revenue over time



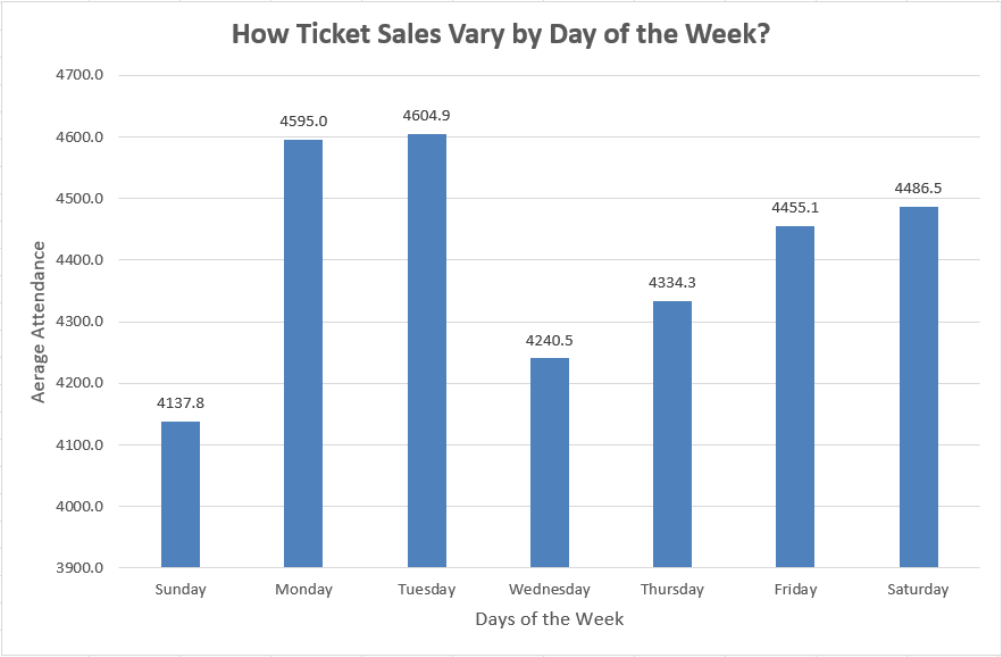
f)

To examine the ticket sales vary by day of the week and month of the year, Attendance feature in Company_X_ShowData dataset is used as the main variable to account the total number of tickets sold per event. Each row of the dataset defines the event with a Show_Date enables to extract the day of the week and month from the date of the show to interpret ticket sales over time.

Then the data is aggregated by Average Attendance per day and month with visualizations to analyze patterns and seasonality in tickets sales.

Ticket Sales vary by day of the Week

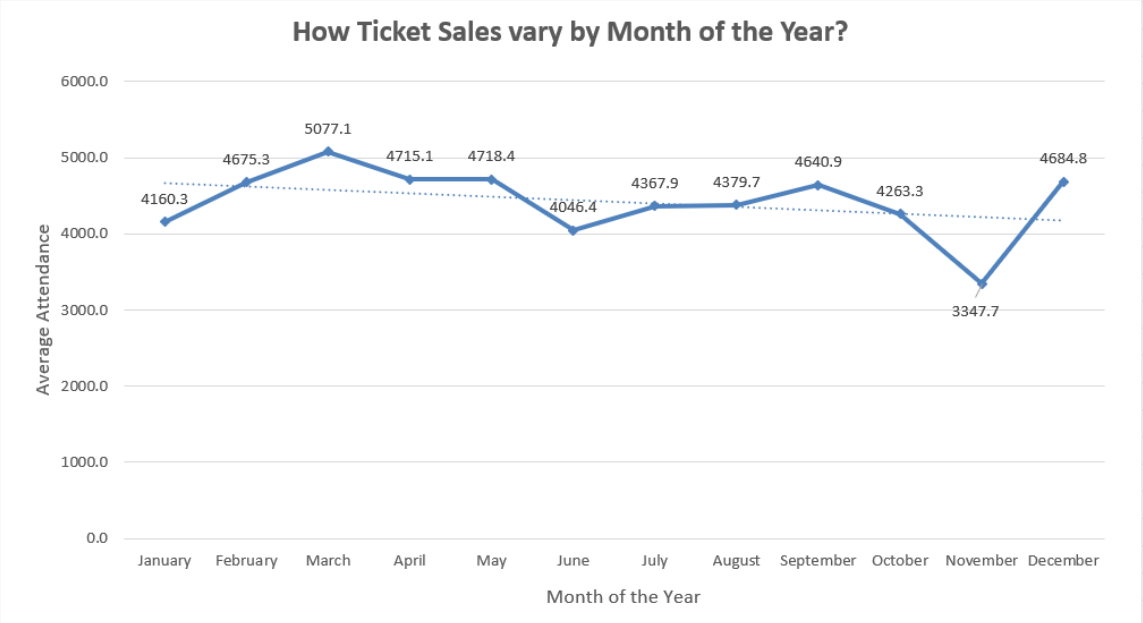
Day of the Week	Average of Attendance
Sunday	4137.8
Monday	4595.0
Tuesday	4604.9
Wednesday	4240.5
Thursday	4334.3
Friday	4455.1
Saturday	4486.5



According to the analysis, Tuesday is observed as the highest attendance with an average of 4604.86. However, Monday has an average of 4595 attendance which is slightly higher than Tuesday. Inferring early weekdays has the highest ticket sales. The lowest attendance was recorded on Sunday which has an average of 4138 displaying a drop in customer turnout at the end of weekend. During weekdays, there is a gradual attendance drop on Wednesdays after peek early week days but higher than on Sunday showing a reduction in engagement in mid-week which decline ticket sales. After that, ticket sales recovers from Thursday to Saturday showing an increasement in attendance but lower than the peak weekdays displaying moderate performance during these days. The pattern infers that ticket sales are not driven in weekends instead has high demand during early weekdays.

Ticket Sales vary by month of the year

Month	Average of Attendance
January	4160.3
February	4675.3
March	5077.1
April	4715.1
May	4718.4
June	4046.4
July	4367.9
August	4379.7
September	4640.9
October	4263.3
November	3347.7
December	4684.8



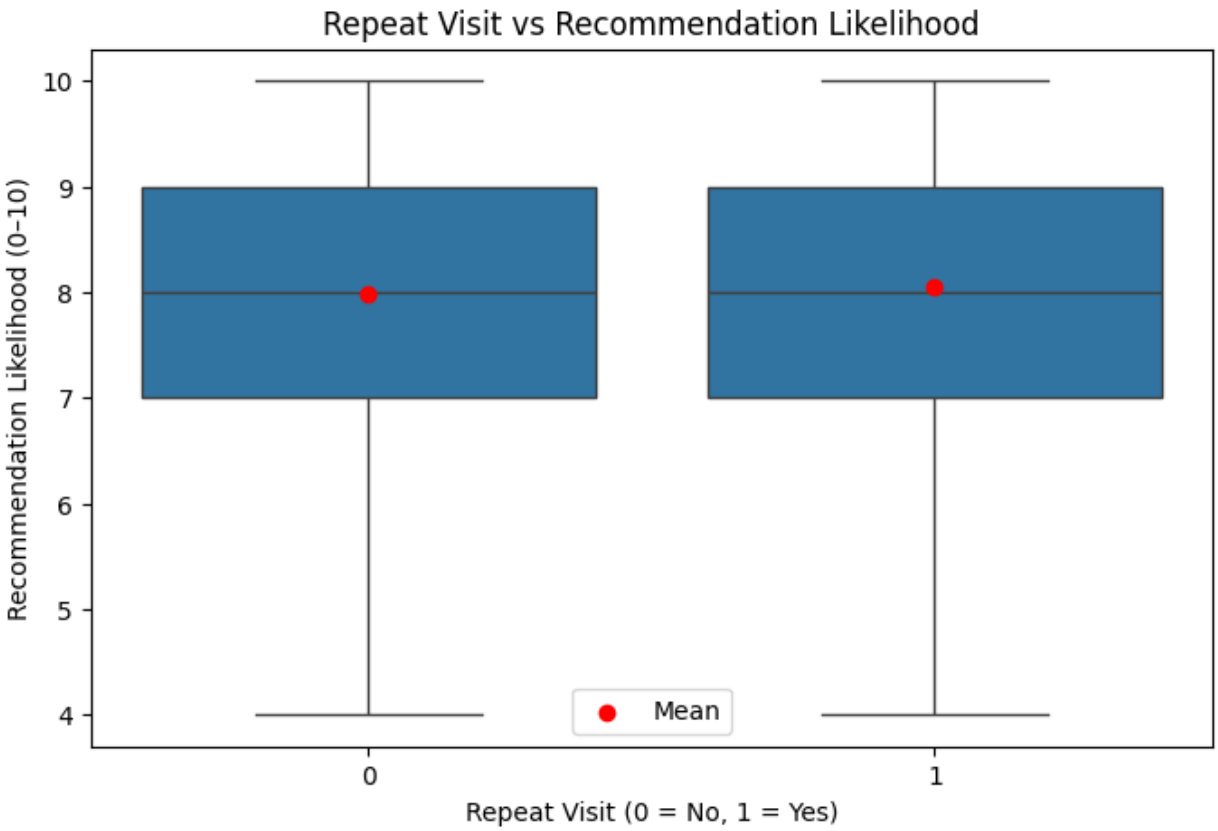
According to analysis, there is no constant attendance across the year showing a seasonality trend in ticket sales but has multi-peak patterns. The highest peak is observed during March with an average attendance of 5077.13 showing a strong demand in this month. In contrast, the lowest attendance was recorded in November with an average of 3347.3 attendance which is a 1700 attendance decline comparing with March signifying a significant drop in ticket sales.

There is a consistent attendance during February to May indicating a strong performance during the early-year period. While the mid-year from June to October shows relatively low attendance comparing with peak month. During this period it shows an off-peak season but in September a secondary peak demand is observed. At the end of the year, December shows a sharp increase in attendance displaying a recovery after the decline in demand in the off-season but its lower than March which had the highest peak. Overall, there are multi-seasonal patterns influence the increasing in attendance and ticket sales.

g)

Key Visualization 1: Comparing Repeat Visits with Recommendation Likelihood.

To compare recommendation likelihood between repeat visits and non-repeat visit a box plot is created to as it effectively visualise components such as distribution, median, spread and outliers to enabling meaningful comparisons between two categories.



The visualization shows that the two box plots overlap almost completely. The Interquartile range in both boxes is approximately between seven to nine displaying an equal distribution with consistent data meaning customer opinion for repeat visit and non-repeat visit is similar. The mean recommendation likelihood for both categories is around eight, indicating both repeat and non-repeat visits have a high tendency to recommend event suggesting that the customers are generally satisfied and willing to recommend their experience but that doesn't mean they will visit again. There is no significant difference between repeat visitors and non-repeat visitors. Therefore, though customer has a strong customer experience does not effectively lead them to return.

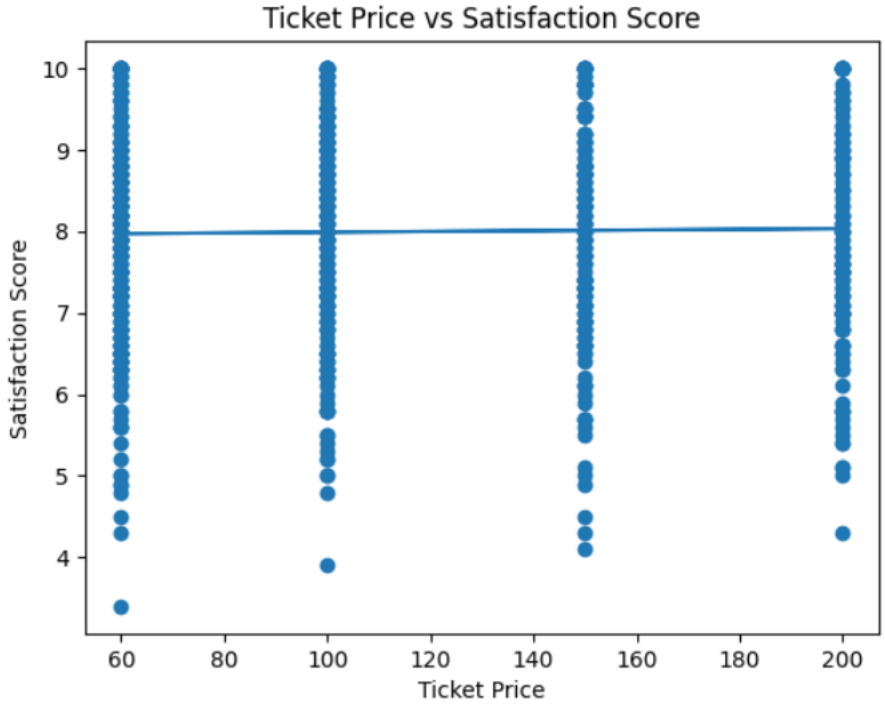
In business context, this shows a clear gap between customer satisfaction and loyalty. Even though, customer report a positive recommendation likelihood doesn't convert to repeat visitors so there is a weak relationship between in customer interest and retention

This implies Company X should focus more on retention strategies rather than improving customer satisfaction by carrying out loyalty programs and targeting customers with incentives

for high recommenders to convert high recommending customer to return customers to boost Company X long term value.

Key Visualization 2: The Ticket Price vs The Satisfaction Score – Insight on Pricing Strategy

Theres a hypothesis that the higher the prices of the tickets, the higher the satisfaction of the customer is due to the premium value that is perceived. This analysis is based on that. Between the Ticket price and the Satisfaction score for examining the relationship, a scatter plot was used. In the visualization, it is indicated that a significant increase in satisfaction score is not there with prices that are higher. Which suggests a relationship that is inconsistent or weak between those two variables. That the customers might not be perceiving additional value in tickets that are expensive is implied by this. So, reconsideration of the pricing strategy should be done by the Company X, by either premium offering enhancement with benefits that are added or maintaining pricing that is competitive for customer satisfaction maximizing and the overall attendance.



Key Visualisation3 : The Seating Region vs The Average Spend – Insight on customer value

The hypothesis that says that the premium seating regions customers contribute higher to the overall revenue through spending that is increased is tested by this analysis. For comparing the seating region with the total spend average, calculated as the total of ticket, the expenditure on drinks and merchandise a bar chart was used. In spending across regions, with segments that are higher tier generating average spend that is greater is a clear variation is shown in the visualization. It is confirmed by this that there are high value customer groups. Company x can implement strategies that are targeted like packages that are premium, loyalty programs, and promotions that are personalized for revenue maximization from these segments which encouraging to spend more for the regions that are lower tier also, based on these insights.

